Mason Phounsavath

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EDUCATION

University of North Carolina - Chapel Hill *Bachelor's, Data Science, Computer Science Minor, Sigma Chi Fraternity* **Aug 2023 - May 2027**

Stanford University
Summer Semester Admit, Computer Science Concentration **Jun 2022 - Aug 2022**

PROFESSIONAL EXPERIENCE

Computer Science and Social Good | *Front End Developer* **Present**

- Collaborated on Full-Stack Web Development - Built a dynamic web application using React.js for front-end development and Firebase Firestore for backend data storage.
- Managed User Authentication & Data Management - Integrated Firebase Firestore to store and manage user profiles with secure database operations.
- Implemented Responsive UI/UX Design - Utilized Tailwind CSS for modern, responsive, and mobile-friendly styling.
- Applied Component-Based Architecture - Designed modular React components for scalability, maintainability, and efficient state management.

Outlier AI | *Machine Learning Engineer* **Jun 2025**

- Designed and reviewed NLP prompts and completions to optimize large language model (LLM) performance.
- Annotated and analyzed model outputs for training and data using Python and machine learning practices.
- Identified edge cases and failure modes to support model fine-tuning and improve AI reliability.

Speedi | *Full Stack Engineer* **May 2025**

- Contributed to a high-growth, campus-wide delivery startup providing UNC students with groceries and dorm essentials in under 15 minutes, leveraging Flutter for mobile development and a PHP backend architecture.
- Assisted in managing internal tools and infrastructure via Microsoft Azure, performance monitoring, and cloud-based scalability.
- Gained experience working across the full software stack while supporting a platform that has generated $10,000+ in early-stage revenue and scaled rapidly across campus.

PROJECTS

Bias Bars: Gender-Based Review Analysis | *Stanford University*

- Natural Language Processing (NLP) & Text Analytics - Extracted and analyzed word frequency distributions in professor reviews, applying tokenization and text normalization techniques.
- Data Processing & Statistical Computing - Implemented Python-based data pipelines to transform raw text into structured insights, leveraging dictionary-based aggregation and frequency normalization.
- Data Visualization & Interactive Dashboards - Developed an interactive Tkinter GUI for dynamic visualization of gender-based language patterns, using Plotlib-style rendering.
- Algorithm Optimization & Style Automation Engineering - Designed efficient data structures and vectorized operations to enhance computational efficiency in large-scale text processing.
- Software Development & Automation - Integrated file I/O operations, batch data parsing, and automated reporting, ensuring robust data ingestion and reproducibility.

2D Dungeon Crawler | UNC Chapel Hill

- Built a 2D dungeon crawler game using Java and JavaFX, implementing both front-end UI and back-end game logic.
- Designed and implemented interfaces and implementation classes to manage game state, including player movement, score tracking, and level progression.
- Integrated Object-Oriented Programming concepts such as encapsulation, interfaces, and polymorphism to structure maintainable code.

Financial Investment vs Performance in ACC Basketball | UNC Chapel Hill

- Built a predictive machine learning model using a decision tree regressor in Python (pandas, scikit-learn) to analyze the correlation between NCAA ACC men's basketball program funding and win percentage.
- Merged multi-source datasets containing team budgets, advanced metrics and performance stats for 15 ACC schools, performing extensive data cleaning, transformation, and visualization using matplotlib and seaborn.
- Achieved an R^2 score of 0.78, validating model accuracy in forecasting 2023 season outcomes based on 2022 financial and performance inputs.
- Provided actionable insights on resource allocation, ROI in athletics, and policy implications related to NIL-era college sports economics, using multiple data visualizations.

Skills

- **Programming Languages**: Python, JavaScript, HTML/CSS, SQL, MySQL, Java
- **Web Development**: React.js, Firebase Firestore, TailwindCSS, Component-Based Architecture
- **Data Science & Analytics**: Natural Language Processing (NLP), Statistical Computing, Data Visualization, Text Analytics, Data Analysis, Machine Learning, Statistical Insights,
- **Data Engineering**: Web Scraping (BeautifulSoup, Requests), Data Parsing, Frequency Normalization, Batch Data Processing, Tableau